The body's defence system

Damaged cells (and white blood cells) secrete increased quantities of histamine, a hormone found in minute quantities throughout the body. Histamine causes blood vessels to dilate, lowers blood pressure and makes the capillary walls more permeable to fluids that need to move out of the small blood vessels to reach the damaged tissue.

1 Define *permeable*.

2 Define *dilate*.

3 Define *secrete*.

The larger diameter of the blood vessels allows more liquid to flow. This in turn causes swelling (due to the increased size of the blood vessels and extra fluid) and tension in stretched skin. The increase in temperature of the inflamed area is due to increased blood flow – blood transfers heat around the body. There may also be an increase in body temperature and pulse rate if a large area is affected. The increased temperature of the body is a defence mechanism to make it harder for pathogens to survive. The swelling may also be painful and throb due to the pulse from connected arteries. Pain is also caused by the release of certain chemicals by the body.

The increased circulation brings special defence agents to the site of cell damage. These include substances to ensure that blood clots in order to stop pathogens from circulating to the rest of the body. *Macrophages* (histiocytes) are specialised types of white blood cells that are attracted to the damaged tissues. They are phagocytes – they surround and digest dead tissue and foreign objects (such as pathogens) that enter the bloodstream. The word comes from the Greek *'phagos'* (one that eats) and *'kytos'* (cell). Phagocytes can pass through the walls of blood vessels.

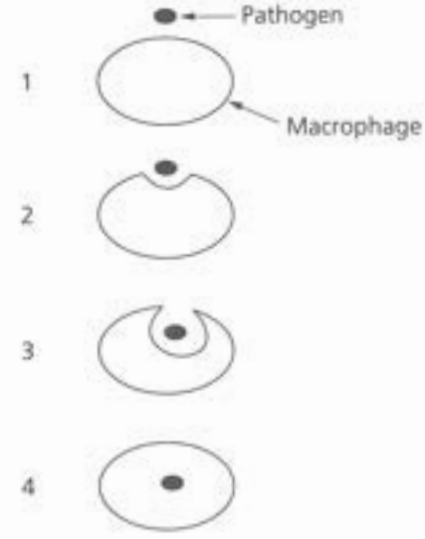
4 The diagram shows a sequence where a macrophage attacks a pathogen. The four stages involved are written below, but in the wrong order. Rearrange the stages in the correct order.

A The cell membrane of the macrophage forms an indentation.

B The macrophage approaches the pathogen.

C The macrophage digests the pathogen and moves on.

D The macrophage engulfs the pathogen.



While inflammation is part of the body's defence mechanisms against pathogens, it has some drawbacks. It can damage tissue and cause pain. If too much histamine is produced, especially in allergic reactions, then people can display painful symptoms. Hay fever and some types of asthma can be caused by over-production of histamines and have to be controlled by drugs called 'antihistamines'. If the inflammation response is overdone, then tissue damage can result. Arthritis is a disease where joints are inflamed.

5 Construct a table showing the advantages and disadvantages of inflammation.

6 Aspirin is often described as an anti-inflammatory drug. What is the likely purpose of anti-inflammatory drugs?

Pain

Pain is an unpleasant experience of the senses that is caused by damage to tissue and detected by sense receptors (special types of nerve cells). These send the message to the brain to tell it that something untoward has happened. As the signal travels from the injury site to the brain, it has to pass across synapses. A synapse is the junction between two neurones, ie the point at which the nervous impulse 'jumps the gap' between them. The brain interprets the signal from the damaged area. Sometimes if pain persists, people take analgesics ('pain killers') to relieve the pain.

7 What is another name for a neurone?

8 Is pain always a bad thing? Explain your answer.

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Cell death can be used to seal off a pathogen. In some cases, a pathogen is completely surrounded by macrophages and lymphocytes so that it is enclosed in a *cyst* (*Greek*, bladder, pouch). The white cells doing this die. Because the pathogen is isolated from its food supply, it also dies. Although body cells die in the process, overall the body benefits.

Identify the terms defined below. Choose from *cyst, lymph, lymphocyte* and *macrophage*.

9 White blood cell, typically found in lymph, that makes antibodies.

10 A capsule or sac with a distinct membrane formed to enclose foreign material.

11 White blood cell that consumes foreign material and protects against disease.

12 Blood that doesn't have red blood cells, platelets or large plasma proteins.

Immunity against disease

The components of this part of the body's defence against infection include:

- **Antibodies** (immunoglobulin): protein produced in lymphocytes as a result of presence of antigens.
- **B-cells** (bone-marrow-derived cell): lymphocytes that secrete antibodies.
- **T-cells**: lymphocyte originating from the thymus gland that reacts with antigens.

Helper T-cells activate killer T-cells and also B-cells, by secreting chemicals that stimulate cloning in T-cells and B-cells. Suppressor T-cells (which only live for a short time) suppress T-cells and B-cells when antigens have been destroyed and there is no longer a need for as many lymphocytes. Memory T-cells remain and when an infection occurs by the same antigen they are reactivated quickly. Helper T-cells stimulate B-cells and T-cells to replicate. In theory, a T-cell could attack a B-cell that had an antigen on its surface. In practice this doesn't normally occur because T-cells are able to recognise individual surface molecules of the millions of different B-cells – just like having a 'fingerprint'. Only genetically identical individuals will have the same surface molecules on their B-cells.

13 Briefly describe, with reference to genetic properties, the types of cells produced by cloning.

14 What is an *antigen*?

15 Give an example of genetically identical humans.

16 Describe the role of helper T-cells.

17 Where do T-cells originate?

Major Histocompatible Complex (MHC)

The body's immune system needs to be able to identify 'self' cells from 'non-self'. With the exception of identical twins, the proteins used for such identification are unique to each individual. *Major histocompatible complex molecules* (MHCs) are protein-based substances that associate with lymphocytes, the T-cells in particular. Their name is derived thus:

- *Major* – important; *Histo* – to do with histamine responses (eg allergic reactions); *Compatible* – well matched, in harmony with something; *Complex* – mixture, compound

The outer membrane of every cell within our body has recognisable MHC markers projecting from it. MHCs act like a 'braille' system for the immune system, able to bind to and identify non-self proteins within a body, by reading the markers. This is why organ transplants can fail. The MHCs quickly recognise the proteins found on the transplanted tissue as non-self and set out to destroy it. The closer the match, coupled with immuno-suppressing drugs (which slow down or stop immune responses), the more likely it is that a transplant will be successful. When the lymphocytes identify the MHC marker, they release the cell and move on. However, if the cell contains any foreign matter (perhaps a bacteria that has attached itself to the cell), despite having the marker, it is targeted for destruction.

18 Suggest the dangers faced by people undergoing immuno-suppressing medications.

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